- Routing algorithms:
  - Random Routing
  - Flow-based Routing
  - Distance Vector Routing
  - Link State Routing
  - Multicast Routing

Routing Strategies

- A large number of routing strategies have evolved over the years.
- Important strategies are:
  - Fixed (Static) routing
  - Flooding
  - Random Routing
  - Flow-based Routing
  - Distance Vector Routing
  - Link State Routing
  - Multicast Routing

Random Routing

- This has the simplicity and robustness of flooding with far less traffic load.
- A node selects only one outgoing path for retransmission of an incoming packet.
- The outgoing link is chosen at random, excluding the link on which the packet arrived.

Random Routing

- A refinement is to assign a probability to each outgoing link and to select the link based on that probability.
- The actual route will typically not be the least-cost route.
  - Network must carry a higher than optimum traffic load.

$$P_i = \frac{R_i}{\sum R_i}$$

Flow-based Routing

- An approach that uses both topology and load information for routing.
- Flow between a pair of nodes is relatively stable and predictable.
- For a given line if the capacity and average flow is known, it is possible to compute the mean packet delay using queuing theory.
- From the mean delays of all the lines, it is easy to calculate the flow-weighted average to get the mean delay for the whole subnet.
- Then the routing algorithm finds the path of minimum average delay.

Fixed

Adaptive Routing

- Routing decisions change as conditions on the network change.
- Two principle conditions that affect routing decisions are:
  - Failure: When a node or trunk fails, it can no longer be used as part of the route.
  - Congestion: When a particular portion of the network become heavily congested, it is desirable to route packets around the area of congestion.
- Classification on the basis of information source: local, adjacent nodes, all nodes.

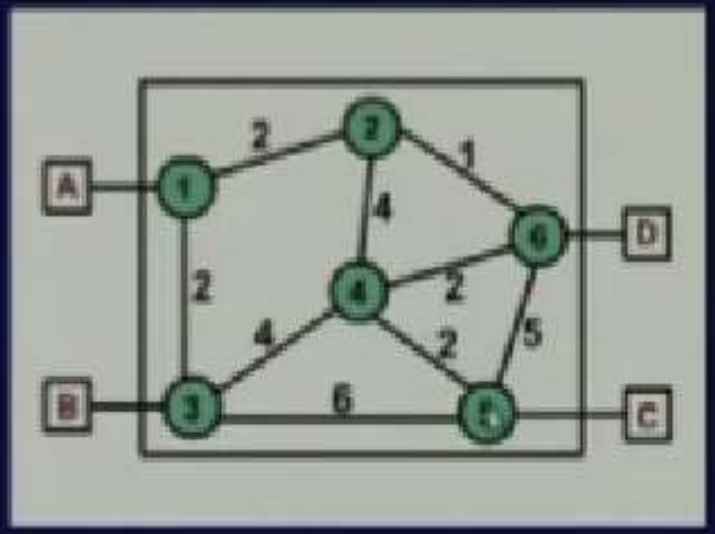
Adaptive Routing

- For adaptive routing to be possible, network state information must be exchanged among the nodes.
- Typical metrics:
  - Number of hops simplest
  - Time delay in msec
  - Total number of packets queued in the path
- How much of information:
  - More information exchange => better routing => more overhead.
  - More frequent => better routing => more overhead.
- Popular approaches:
  - Distance vector routing
  - Link state routing

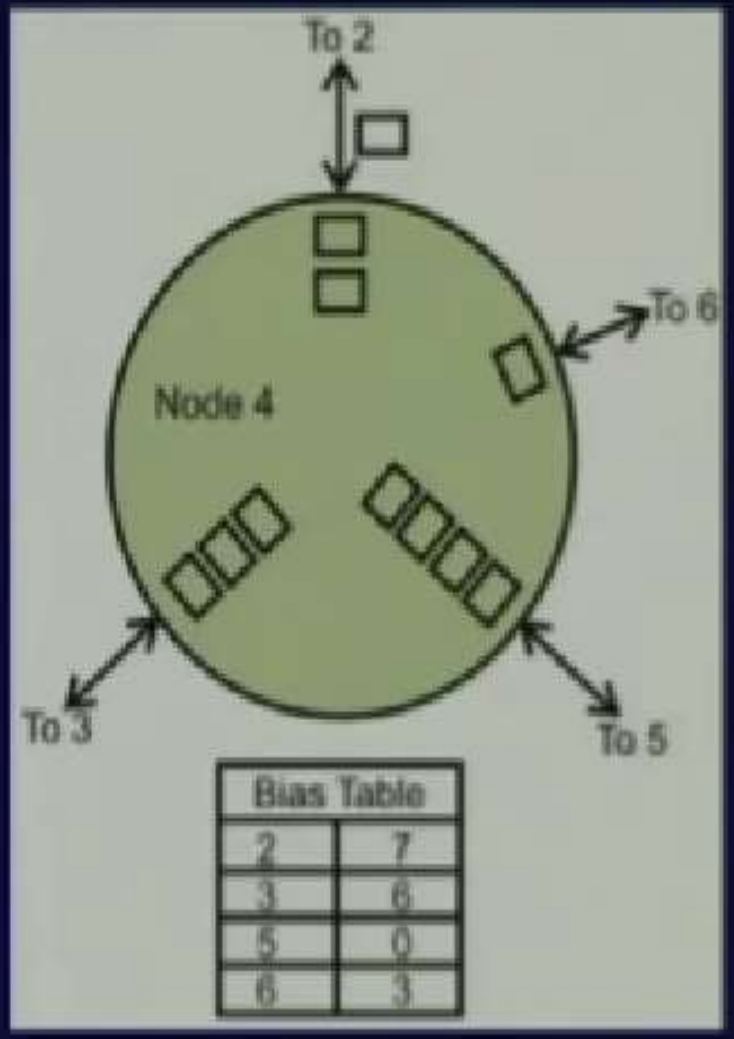
Isolated Adaptive Routing

- A node routes an incoming packet to the outgoing link with the shortest queue length Q.
  - Balances load on outgoing links.
  - The chosen outgoing link may not be heading in the right direction.
- To take direction into account, each link emanating from the node has a bias B_i, for each destination *i*.
  - For each arriving packet heading for node *i*, the node would choose the outgoing link that minimizes $Q+B_i$.
  - A node would tend to send packets in the right direction, with a concession made to current traffic delays.

Isolated Adaptive Routing: an example



Packet arrives from node 2 with destination address of node 5



Bias Table	
2	7
3	6
5	0
6	3

Distance Vector Routing

- Key characteristics:
  - Knowledge about the entire network
  - Routing only to neighbours
  - Information sharing at regular interval

- Each node maintains a routing table having one entry for each node with two other fields; preferred next node and an cost estimate of the distance, based on one of the metrics.

Destination next hop

Network ID	Cost	Next Router

Link State Routing

- Basic steps:
  - Identify the neighboring nodes
  - Measure the delay or cost to each of its neighbors
  - Form a packet containing all the information
  - Send the packet to all other nodes (flooding)
  - Compute the shortest path to every other node (Dijkstra'a algorithm)

Link State Routing

- Key characteristics:
  - Knowledge about the neighborhood
  - Routing to all flooding
  - Information sharing at regular interval

who Link State Packet next hop

Advertiser ID	Network ID	Cost	Next Router

Link State Versus Distance Vector

- Link state algorithms converge more quickly and are somewhat less prone to routing loops than distance vector algorithms.
- Link-state algorithms require more CPU power and memory than distance vector algorithms. Link-state algorithms, therefore, can be more expensive to implement and support.
- Link-state protocols are generally more scalable than distance vector protocols.

Drawbacks of Adaptive Routing

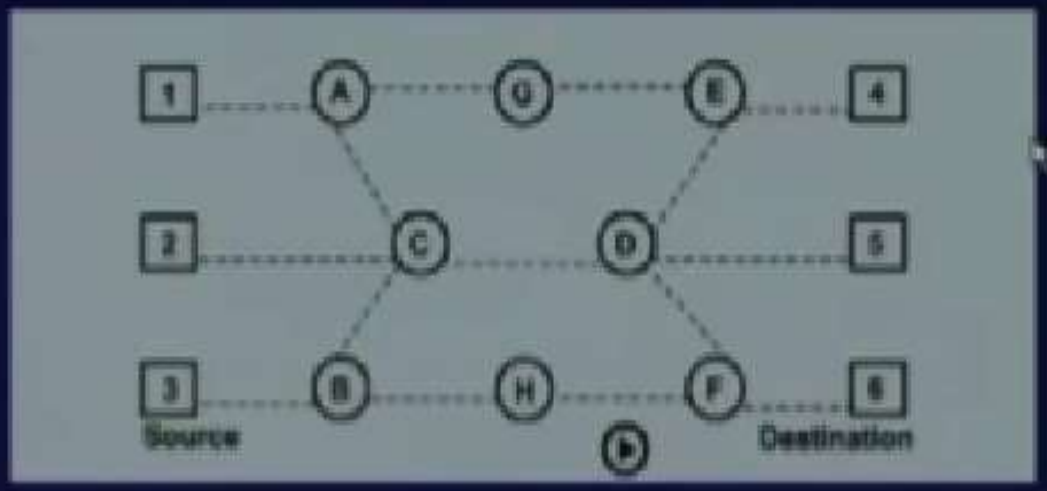
- Routing decision is more complex => more processing burden on the switching nodes.
- Depends on status information that is collected at one place but used at another => traffic overhead increases.
- It may react too quickly to changing network state, thereby producing congestion-producing oscillation.

Thrashing

- Despite the drawbacks, adaptive routing is widely used.
  - Improves performance
  - Can aid in congestion control

Multicast Routing

- Sending a packet to a selected group is known as **multicasting.**
- To do multicast routing each node computes a spanning tree covering all other nodes in the subnet
- When a packet is sent, the first router examines its spanning tree and prunes it, removing all nodes that do not lead to stations, who are member of the group



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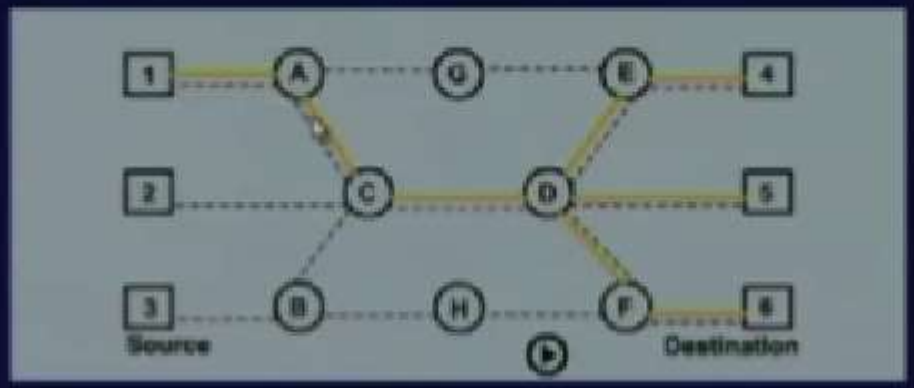

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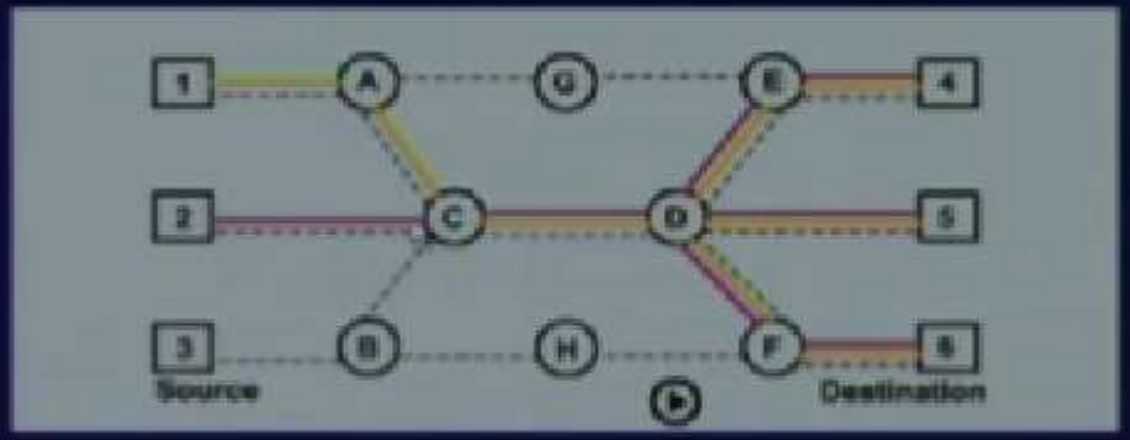

1. Why routing is important in a packet-switched network?

Ans: The path to be followed by a packet introduced in a packet switched network is decided by the routing algorithm. Routing tries to find out the least-cost or the optimized path between the source and the destination stations. If routing is not done properly, delay increases an this may lead to congestion.

2. What are the primary conditions that affect routing?

Ans: The primary conditions that affect routing are:

Failure: When a node or trunk fails, it can no longer be used as part of the route.

Congestion: When a particular portion of the network become heavily congested, it is desirable to route packets around the area of congestion.

3. What is flooding? Why flooding technique is not commonly used for routing?

Ans: Flooding is one type of non-adaptive routing technique where no network information is used. In case of flooding as each node receives a packet, it is re-transmitted or forwarded to all the links connected to the node (except the link through which the packet has arrived).

- Flooding is not commonly used for routing for the following reasons:
  - Flooding leads to unbounded number of packets
  - May lead to congestion in the network
  - A number of copies of the same packet is delivered to the destination node

4. In what situation flooding is most appropriate? How the drawbacks of flooding can be minimized?

Ans: Flooding is most appropriate in some critical operations, like military network, because of its robustness. In flooding routing technique the packet delivery is guaranteed if a path exists.

- The drawbacks of flooding can be minimized by the following two ways:
  - While forwarding a packet each node should find whether the particular packet has been already transmitted. If so, the second transmission of the packet should be stopped.
  - Hop-count information should be maintained at each node. A packet is not forwarded, if hop-count is more than the specified limit.